

Arduino-based Desk Lamp System Design with Sitting Posture Monitoring

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Abstract—The radiation shielding system for wireless charging of substation inspection robots is designed to reduce or eliminate the electromagnetic radiation generated by the robot during wireless charging. This system plays a crucial role in ensuring the proper functioning of inspection robots and maintaining human health. Inspection robots in substations are usually based on the principle of electromagnetic induction for the transfer of electrical energy in wireless charging systems. Such a charging method releases electromagnetic radiation, which may adversely affect nearby equipment and the human body. With the radiation shielding system, we are able to reduce or completely eliminate electromagnetic radiation, thus ensuring the safety and stability of the operation of the inspection robot in the substation when it is charging wirelessly.

Keywords—component; Sitting posture detection, Arduino, Sitting posture monitoring system, Multi-functional intelligent Desk lamp

I. INTRODUCTION

Nearsightedness is becoming more and more common among young people, and the key factors are incorrect sitting posture and heavy school load. Poor sitting posture not only damages the eyes, but also predisposes to lumbar spine disorders[1-3]. In view of the impact of poor sitting posture on adolescents' academic performance and quality of life, it is necessary to design a sitting posture monitoring system suitable for young people.

Currently, there are two main methods for recognizing human sitting posture: one is a sensor-based method, which uses various types of sensors to capture changes in sitting posture and process the data to determine whether it is a normal sitting posture; and the other is a visual image processing method, which uses a high-resolution camera to collect images of sitting postures at different moments for identification. Wongpatikaseree Konlakorn's team designed a working environment sitting posture recognition system[4], which adopts a visual recognition method to analyze the sitting posture by acquiring images and building a skeleton model through the Kinect camera, but it has limitations such as inconspicuous recognition posture and low recognition rate. Shao-Hua Liu designs a pressure sensor-based sitting posture monitoring system[5], which analyzes the pressure distribution

data to determine the classification of sitting posture. Mingyang Sun builds a sitting posture monitoring robot[6], which utilizes thin-film pressure sensors to monitor, but there are shortcomings such as needing to touch and reach the pressure threshold to judge.

Currently, research results and breakthroughs have been made in sensor-based and visual image processing-based sitting posture detection methods, but there are still some problems in correctly recognizing sitting postures. In visual image processing: the process of collecting pictures needs to ensure the brightness of the environment, data processing needs to build a more complex model, the accuracy of image recognition is not well guaranteed. In the use of sensors: it is convenient to collect data about the user's sitting posture, but the accuracy of the collected data is not high. When the number of sensors is too large, the cost will also increase. Pressure sensors need to be in contact with people, and the user's experience will be greatly reduced. Therefore, this paper designs a multifunctional desk lamp lighting system with sitting posture monitoring, which uses fewer sensors and does not require contact with the human body for the sitting posture recognition module.

II. SITUATION ANALYSIS AND OVERALL SYSTEM DESIGN

A. Sitting Analysis

When people sit, the muscle strength of the legs and abdomen will gradually decrease or disappear, thus promoting the blood flow for the body to relax and rest. From the perspective of human anatomy, the most weight-bearing part is the sciatic area, while the lower thigh contains a large number of nerves and the aorta, and can not withstand excessive pressure, otherwise it will impede blood circulation, affecting the nerves, resulting in localized numbness[7]. A standard sitting posture should ensure an even distribution of pressure on the cervical joints and their attaching muscles, and therefore needs to be set as the goal of the control system.

B. Overall System Framework

The overall structure of the system in this paper is mainly composed of four main parts: sitting posture acquisition and monitoring module, clock display module, temperature and

humidity module, and desk lamp control module. The sitting posture module contains components such as data acquisition circuit, ultrasonic sensor, OLED screen, voice reminder, and led small lamp, and most of the other modules are composed of individual sensors[8]. The main flow chart of the multifunctional intelligent desk lamp system is shown in Fig 1:

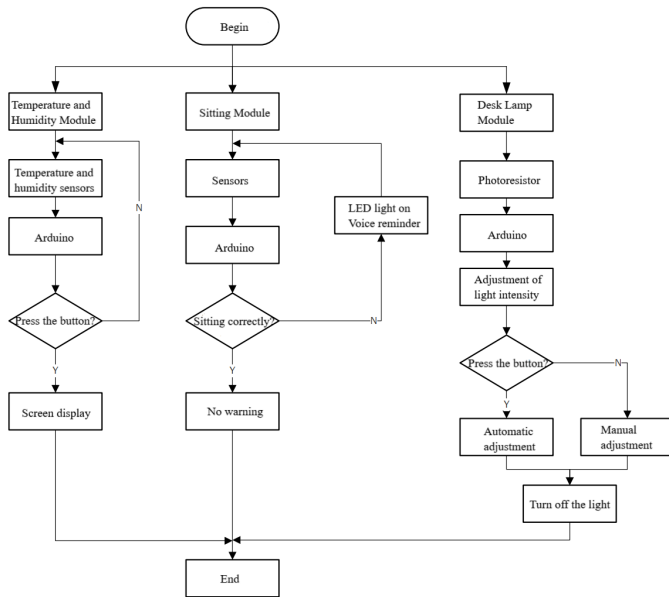


Figure 1 Main flow chart of the system

III. HARDWARE ARCHITECTURE DESIGN

A development board is a circuit board that integrates a microcontroller with other peripherals. It is composed of components such as microcontroller, power supply, various types of ports, sensors, etc. It has various types of standardized interface protocols, which facilitates secondary development by users. To facilitate the construction of various system circuits. The system designed in this paper selects the Arduino series development board to build. It has more development resources, simple operation and stable hardware. Arduino series contains two parts, hardware and software, the development board is its hardware platform, in terms of software, through the Arduino IDE software for program writing.

For the most important detection module, this paper uses an ultrasonic sensor to monitor the user. It sends a short pulse of high-frequency acoustic signal into the air, and when it encounters an object, the signal will be reflected back to the receiver of the ultrasonic sensor. The sensor then converts the received reflected signal into an electrical signal. The control circuit inside the ultrasonic sensor calculates the time it takes from sending the signal to receiving it, and then the distance between the object and the sensor is calculated from the time. Where the workflow is shown in Fig 2, when the sitting monitoring state is turned on, if the user is in an incorrect sitting position, the MP3 voice reminder module is activated, and after the user corrects the sitting position, the monitoring returns to the normal state.

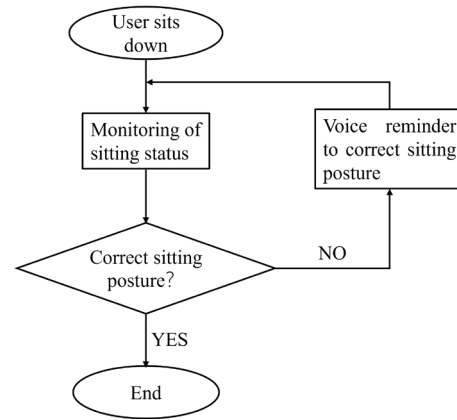


Figure 2 Flowchart of Sitting Posture Detection

Since the monitoring of sitting posture requires a voice reminder of incorrect sitting posture, a voice broadcast module needs to be designed. The voice playback module based on MP3 decoder chip is used for voice prompts, voice announcements and other functions. With a variety of playback modes and control methods, it can be controlled by serial port, IO port, touch switch and other ways.

In addition, the overall system also includes some other modules. For example, the temperature and humidity module is used to detect the environment around the user. The sensing element quickly and accurately measures the ambient temperature and humidity, and converts the measurement results into a digital signal output through the serial interface to the processed digital signal output to the external device (Arduino Mega 2560 development board) for communication, to achieve data transmission and processing. In order to realize the control system to automatically adjust the light intensity function, selected the photoresistor sensor to achieve automatic dimming, the sensor is suitable for Arduino, with VCC, GND, IN port, through the IN port to obtain the ambient light intensity and convert analog into digital uploaded to the main controller. DS1302 clock module is selected to display the time in real time on the OLED display. It adopts SPI protocol with low power consumption and automatic power switching function, which can keep the time and date information during power failure.

IV. SOFTWARE ARCHITECTURE DESIGN

A. Software Platform

Arduino IDE (Arduino Integrated Development Environment) software is an open-source integrated development environment for writing and uploading code for Arduino development board, and its programming language is developed based on C language. In this paper, the Arduino Mega 2560 development board is used, so to realize the various functions of the multifunctional smart desk lamp, we need to program and design the development board through the Arduino IDE software.

B. Programming Framework Design for the System

The main function programs of the system designed in this paper are: brightness adjustment program, OLED display

program, sitting posture monitoring program, clock module program, temperature and humidity module program and MP3 voice program. As shown in the figure for the software programming logic structure diagram. After editing the functional program, it is also necessary to realize the control of the system through the keyboard keys. Fig. 3 illustrates the program functions that the system has.

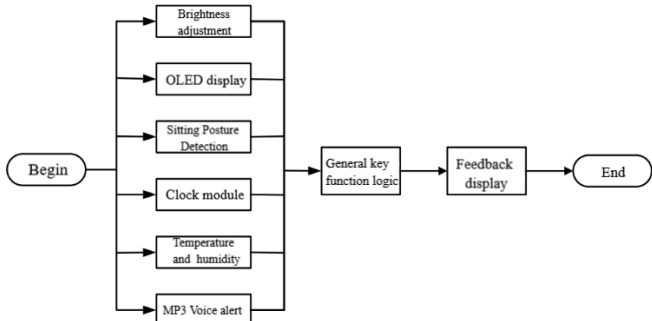


Figure 3 Logical structure of software programming

C. Programming of the Sitting Posture Monitoring Module

In the system design, the most important module is the sitting posture monitoring, and the ultrasonic sensor is the detection device of this module. First of all, we need to check whether the ultrasonic sensors are able to collect data properly. In the sitting posture detection module, four ultrasonic sensors are used to monitor the sitting posture of the user's front chest, back, left side and right side of the body respectively, and before collecting the sitting posture data, the program design of this module is needed[9-10]. The program of this module mainly defines the pins of the four sensors and then creates the functions for the front chest, back, left side of the body and right side of the body respectively. First set up a pulse-driven sensor to monitor the sitting posture, and then control whether to collect this data through the keypad. If the data of the correct sitting posture is collected, a reasonable monitoring range is set on the basis of this data, and when the user sits lower than this range, the MP3 voice module starts to work, and at the same time, the corresponding small LED light will also light up.

V. TESTING AND ANALYSIS

A. Sitting Posture Monitoring Module Test

After installing the device module in place, the user is required to sit on the chair to ensure that each sensor measures the data in the appropriate position and use the matrix keyboard keystrokes to record the user's correct sitting posture data. At this point the user can test whether the system is working properly by changing his/her sitting posture, for example, leaning forward, leaning back, leaning left, leaning right etc. When the user is in a bad sitting posture, the small light of the corresponding sitting posture will light up, and the MP3 voice module will also send out voice reminders to remind the user to adjust the posture. The user data collected in the test is shown in the Table 1.

TABLE 1 USER SITTING DATA

	Chest distance (cm)	Back distance (cm)	Distance left (cm)	Distance right (cm)
User 1	23	24	14	16
User 2	24	19	12	16
User 3	31	18	14	15
User 4	29	23	12	10

Based on the analysis of the data in the table above, factors such as height and weight of each individual will produce a large variability in the data under the same position. This also illustrates the need to take into account the effect of individual variability when performing correct sitting posture data measurements. However, despite these variabilities, the measurement of correct sitting posture data is still very important. This is because this data can provide the system with key information that can be programmed into the software to set reasonable sitting ranges, thus reducing the potential for error.

B. Temperature and Humidity Module Testing

In the designed system, the role of the temperature and humidity module test is to verify the accuracy and precision of the temperature and humidity module, to ensure that it can correctly measure temperature and humidity, and provide reliable data output, and the purpose of this test is to detect the environment in which the user is located, to ensure that the user's comfort. After the test is completed, the data output from the temperature and humidity module is checked to determine whether it is within the acceptable range to prevent the data measured by the module from having too large an error. As shown in Fig. 4, it is the specific data of the temperature and humidity of the environment in which the user is collected.

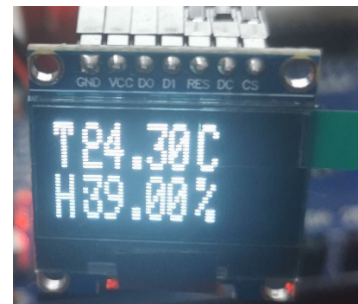


Figure 4 Temperature and humidity data display

C. Desk Lamp Lighting Module Test

The brightness of the desk lamp can directly affect our vision health and work efficiency. In order to meet the needs of different users, two modes of manual brightness adjustment and automatic brightness adjustment are set up in the desk lamp module so that users can adjust the brightness according to their actual situation. In manual brightness adjustment mode, users can adjust the brightness of the light through different keys. The automatic brightness adjustment mode automatically adjusts the brightness of the desk lamp according to the ambient brightness so that users do not need to adjust it

manually, which is more intelligent and convenient. In order to realize these two modes, the desk lamp module uses an 8-bit LED cluster and a photoresistor to control the brightness. The photoresistor can sense the brightness of the surrounding environment and pass the perceived signal to the control circuit to automatically adjust the light brightness, while manual adjustment of brightness is realized by controlling the brightness of the LED light set through the control matrix keyboard to realize the user's manual adjustment. The brightness of the desk lamp can be adjusted through the matrix keypad. It has multiple brightness levels up to 25 levels of brightness. The overall hardware assembly diagram of the desk lamp module is shown in Fig. 5, which consists of several electronic components, among which the LED light set and the photoresistor are the core components to realize the brightness control.



Figure 5 Hardware diagram of table lamp module

VI. CONCLUSIONS

This paper designs a multifunctional intelligent desk lamp lighting system with sitting posture monitoring. It meets the needs of users when they work and study, and at the same time, it can monitor their sitting postures in time to help users avoid health problems caused by sitting postures. Its main functions are sitting posture monitoring, clock display, desk lamp brightness adjustment and voice reminder. This system uses the Arduino Mega 2560 development board as the main controller, combined with a variety of sensors to realize the construction of multifunctional intelligent system. The original master controller and sensors used in this paper are described in detail. In addition, practical operation tests were conducted to carry out the experiment of each function of the multifunctional intelligent desk lamp system. Eventually realized the overall system can be timely monitoring and reminding according to the user to make different sitting movements. The overall system can provide users with early warning and protection, correct bad sitting posture, and help improve the efficiency of work and study.

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